

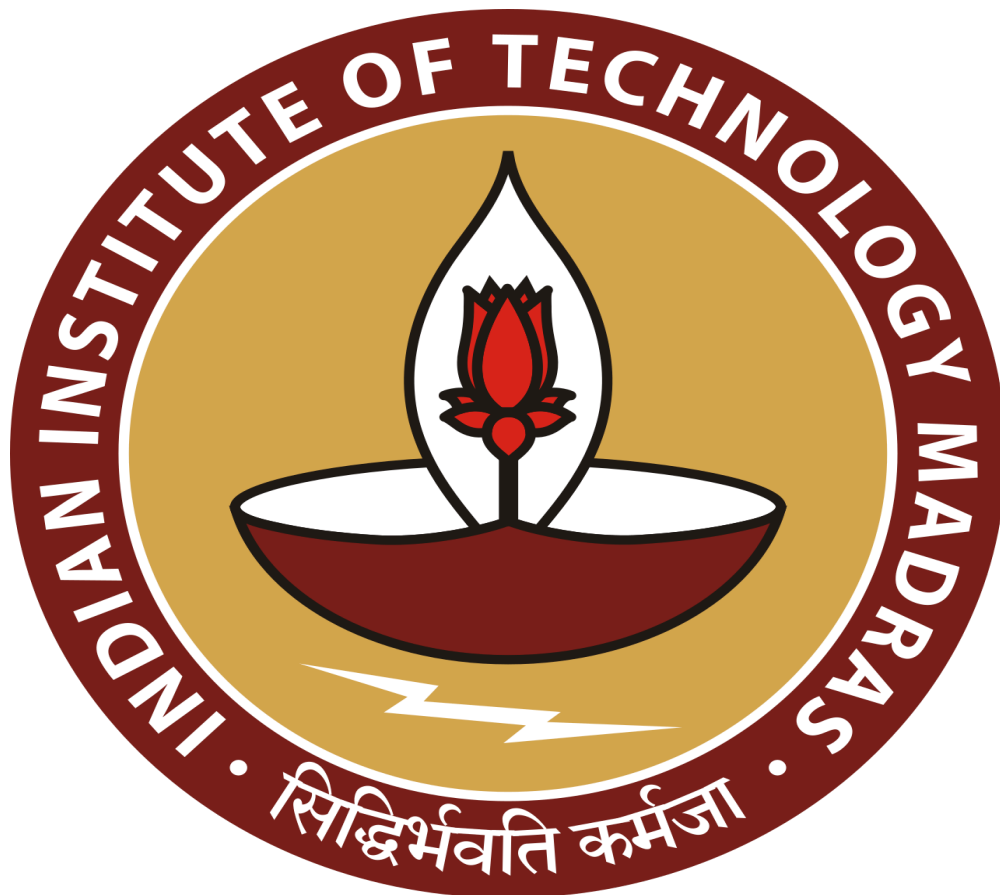
An Analytical Study of Demand Patterns and Price Ambiguity in a Construction Materials Retail Store

A Proposal report for BDM capstone Project

Submitted by

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Declaration Statement

I am working on a project titled “An Analytical Study of Demand Patterns and Pricing Ambiguity in a Construction Materials Retail Store.” I extend my appreciation to **Trishul Cement and Hardware Store**, for providing the necessary resources and information that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analysed to assure its reliability.

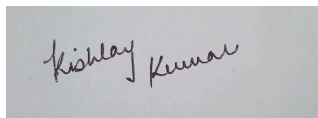
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:

A rectangular box containing a handwritten signature in black ink. The signature is written in a cursive style and reads "Kishlay Kumar".

Name: Kishlay Kumar

Date: 13/03/2026

Executive Summary

This project focuses on the analysis of a small construction material store that deals with both B2B and B2C customers with the sale of products such as cement , thermo-mechanically treated (TMT) bars , stirrups , coating materials such as paints and primers. The business serves mainly individual customers who account for 90% of the sales and remaining 10% consists of small contractors .

The major business challenges faced by store are inventory management , fluctuation in demand, ambiguity in pricing and profitability structure . Due to variations in supplier prices and informal market pricing (mainly for cement, TMT bars and stirrups) , the store often faces uncertainty in determining appropriate selling prices and profit margins which can affect the overall profitability of the business.

To tackle these challenges at ‘Trishul Cement and Hardware Store’ firstly, I will explore available raw data and apply data-driven techniques to transform it into structured and meaningful insights . The analysis will focus on understanding how demand changes over time, how pricing decisions affect profitability, and how inventory is managed so that the store can make better decisions and can enhance overall business profitability.

Organization Background

Business Name: Trishu Cement and Hardware Store

Address: Opposite friend’s Colony , Jaganpura Road , Patna , Bihar

Owner’s Name: Shreekant Kumar

The business I have selected for this project is Trishul Cement and Hardware Store , a local Construction materials retail shop . The store was established in June 2000 with an initial

investment of Rs 20000 with the aim of supplying construction materials to customers in the surrounding area .

Problem Statement

The problem statement that could be identified after conversation with business owner is as follows -

- **Demand Fluctuation**

Explanation - Customer Demand for Construction materials varies across seasons, making it difficult to predict sales.

- **Ambiguity in Pricing and Profitability**

Explanation - The Business sometimes sets higher selling prices to maintain profit ,

However due to strong competition in the local market , customers choose to purchase materials from other nearby stores .This leads to decline in sales leaving the store with Unsold inventory .

- **Inventory Management Challenge**

Explanation - The Business faces difficulty in maintaining the right inventory for different construction materials such as Cement , TMT bars and stirrups due to unpredictable demand .

Background of the problem

During the initial stage of this project, I spoke with the owner of the business to understand the challenges faced in running the business. The owner explained that sometimes the store sets

slightly higher selling prices to maintain profit margins. However, due to strong competition from nearby stores, customers may choose to buy materials from other businesses offering lower prices.

When this happens, some materials remain unsold in the store's inventory. This can be a serious issue for certain construction materials. For example, if cement bags remain unsold for a long time, exposure to moisture can cause them to harden and become unusable. Similar situations can occur with TMT bars and stirrups, which may start to rust if they remain in storage for longer periods. Situations like this can lead to financial losses for the store.

Problem Solving Approach

The project will use a combination of quantitative and qualitative methods to understand the business challenges.

Qualitative Method : This will involve talking to the Business Owner to gather qualitative information about the store's operations, challenges and decision making practices. Also, I will be observing nearby businesses to understand how pricing and inventory are managed in the local market.

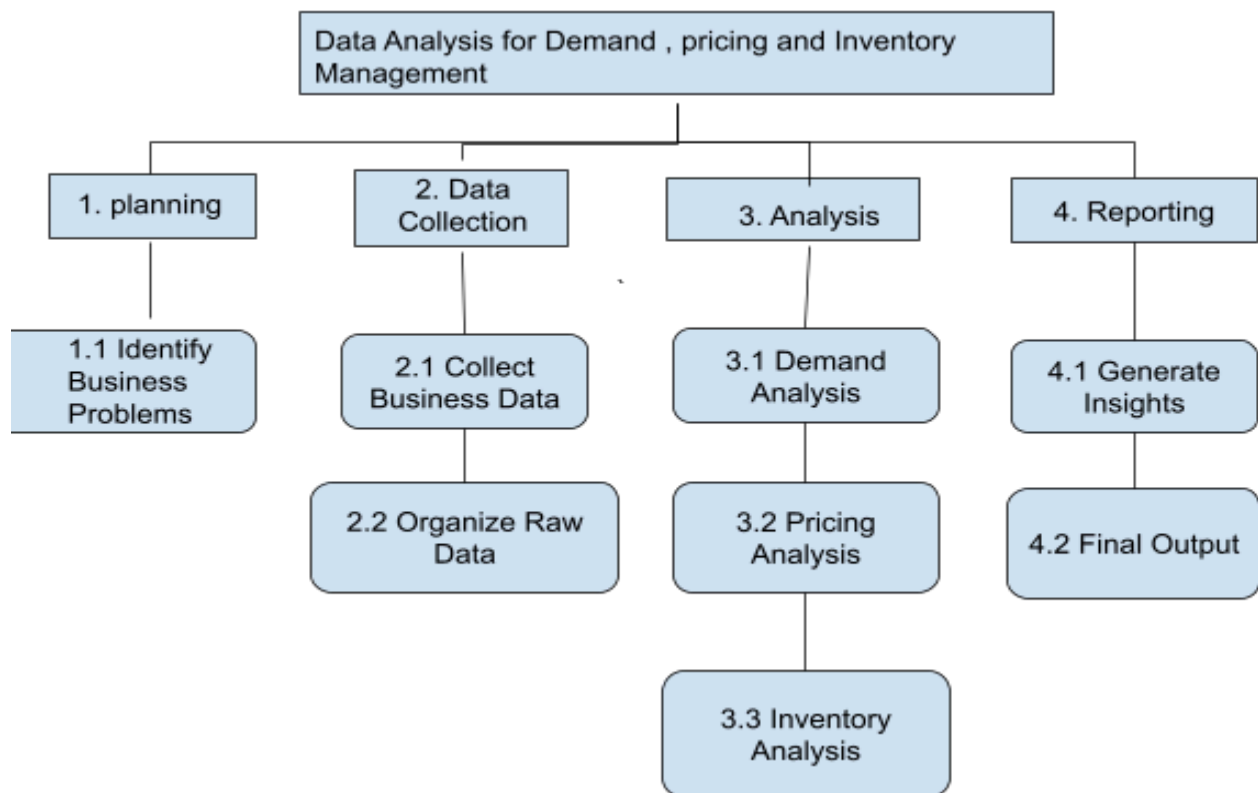
Quantitative Method: This will involve collecting 6 months raw data. The collected data will be cleaned and organized in a structured format. The structured data will include variables such as Product Type(Cement, TMT bars, stirrups of different shapes, Quantity purchased). These variables are relevant to problem statements because they help analyze demand patterns, pricing decisions and inventory management. Tools such as Microsoft Excel and Google Sheets will be used to explore the data more comprehensively and gain necessary insights. Visualization tools such as bar charts, line charts, and pie charts will be used to better understand trends in sales, product demand, and inventory levels. In addition to organizing the data, basic analytical techniques will be applied to identify patterns and relationships within the dataset. For example, demand trends for different construction materials will be examined over time to understand how

customer requirements change across different periods. Comparing the sales of different products will also help identify which materials contribute the most to overall sales.

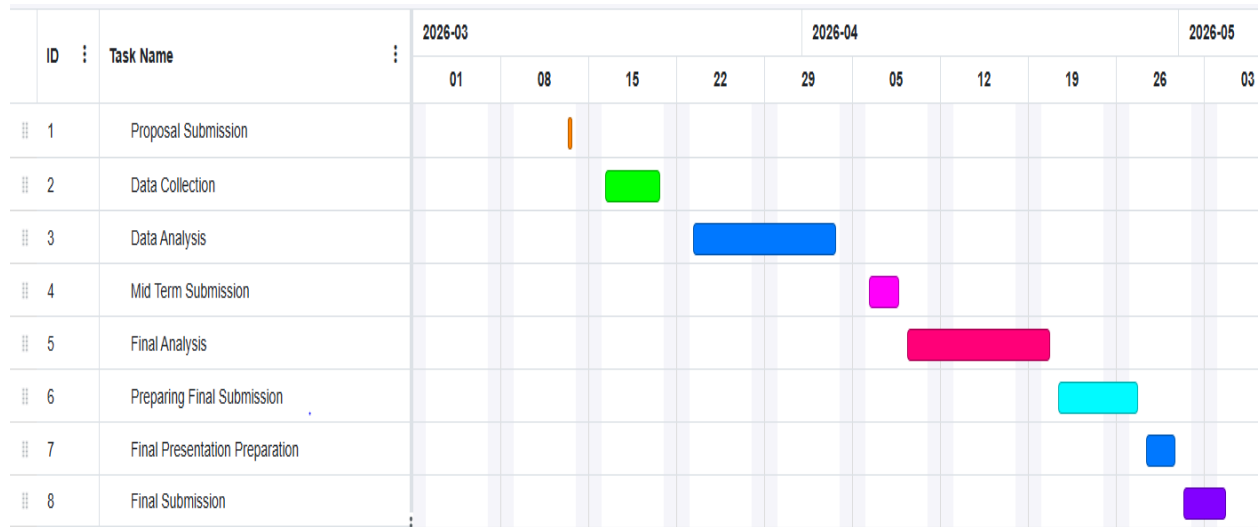
Furthermore, an inventory aging analysis will be conducted to examine how long certain materials remain in stock before being sold. This will help identify slow-moving products and highlight potential risks associated with long storage periods, such as cement hardening due to moisture or TMT bars developing rust. By combining these analytical techniques with insights gathered from the business owner and observations of nearby businesses, the study aims to generate meaningful insights that can support better pricing decisions and more efficient inventory management.

Expected Timeline

Work Breakdown Structure



Gantt Chart



Expected Outcomes

The expected outcome of this project is to develop a better understanding of the store's operations and identify ways to improve its efficiency. By examining the sales and inventory data collected from the business, the study will aim to identify patterns in customer demand for different construction materials. Understanding these patterns can help the store make more informed decisions about purchasing materials and maintaining appropriate stock levels.

The study will also look at how pricing decisions influence customer buying behavior. This will help the store determine whether higher prices are affecting sales and whether adjustments may be needed to remain competitive with nearby businesses.

In addition, the project will analyze how long different materials remain in inventory. This can help highlight potential risks, such as cement hardening due to moisture or TMT bars developing rust when stored for extended periods. Overall, the study aims to generate practical insights that can support better pricing decisions and more efficient inventory management for the store.